

Matrix and System of Equations

*Author: Jaehoon Song**Release: 2025-05-02 20:13:27-04:00***1. Matrix Equation by Inverse Matrix** (9 points)

Use the inverse matrix method to solve the following systems of equations.

- (a) Solve for the matrix
- X
- in the equation shown. (3 points)

$$X \begin{bmatrix} -1 & 4 \\ 2 & 9 \end{bmatrix} = \begin{bmatrix} -13 & -50 \\ -8 & -19 \end{bmatrix}.$$

Answer: your answer here.

- (b) Solve for the matrix
- X
- in the equation shown. (3 points)

$$X \begin{bmatrix} 6 & 1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 14 & -2 \end{bmatrix}$$

Answer: your answer here.

- (c) Solve for the matrix
- X
- in the equation shown. (3 points)

$$X \begin{bmatrix} 1 & -2 \\ 4 & -9 \end{bmatrix} = \begin{bmatrix} -5 & 12 \\ 9 & -20 \\ 4 & -8 \end{bmatrix}.$$

Answer: your answer here.

2. **Matrix Equation by RREF** (9 points)

Solve the following systems of equations by writing a matrix to represent each system and reducing it to **reduced row echelon form (RREF)**. Include the row operations you use for decomposition.

(a) Solve the system of equations. (3 points)

$$\begin{aligned}2x + 4k - z &= 15, \\2z + k &= -2, \\4z + 3y - 3k &= -9, \\2z - y &= -11.\end{aligned}$$

Answer: your answer here.



(b) Solve the system of equations. (3 points)

$$\begin{aligned}2x + y &= 2, \\y + z + 4k &= 21, \\2k - z &= 13, \\2z - y &= -10.\end{aligned}$$

Answer: your answer here.



(c) Solve the system of equations. (3 points)

$$\begin{aligned}2x - 8z + k &= 14, \\-4z + 2y - 2 &= 0, \\4y - 9k &= -12, \\x + k - z &= 1.\end{aligned}$$

Answer: your answer here.



3. **Matrix Equation from System of Equations** (4 points)

Minju, Jiya, Hanna, and Jackie are ordering candy and flower arrangements for their mothers for Mother's Day. Each box of candy has a fixed cost, and each flower arrangement has a fixed cost, but the website they are ordering from does not advertise the price. Jackie's mom is allergic to flowers and Hanna's mom does not like candy, so they only want to order something their mom likes. To figure out the cost, Minju and Jiya order first. Minju orders 3 boxes of candy and 4 flower arrangements, and her total is \$38. Jiya orders 12 boxes of candy and 2 flower arrangements, and her total is \$40. Hanna and Jackie each have \$45 to spend. Use this information to answer the following questions:

- (a) What is the cost per unit for flower arrangements? (1 point)

Answer: your answer here.



- (b) What is the cost per unit for boxes of candy? (1 point)

Answer: your answer here.



- (c) How many boxes of candy can Jackie buy? (1 point)

Answer: your answer here.



- (d) How many flower arrangements can Hanna buy? (1 point)

Answer: your answer here.



4. **Matrix Transformation (Points)** (11 points)

Answer the following questions about transformations of triangles:

- (a) A triangle has vertices $A(1, 2)$, $B(3, 4)$, and $C(5, 6)$. The triangle is then rotated 90° counterclockwise around the origin.

- i. Find the coordinates of A' , the image of vertex A after the rotation. (1 point)

Answer: your answer here.



- ii. Find the coordinates of B' , the image of vertex B after the rotation. (1 point)

Answer: your answer here.



- iii. Find the coordinates of C' , the image of vertex C after the rotation. (1 point)

Answer: your answer here.



- iv. Find the area of triangle $A'B'C'$. (1 point)

Answer: your answer here.



- (b) $\triangle ABC$ has vertices $A(a, 2)$, $B(-1, 3)$, and $C(6, a)$. It is known that $\triangle ABC$ has an area of 6 square units.

- i. Find all possible values of a . Show the work that leads to your answer. (3 points)

Answer: your answer here.



- ii. If $\triangle ABC$ is rotated 15° clockwise around the origin, find the coordinates of B' , the image of vertex B after the rotation. (4 points)

(Hint: $\cos(a \pm b) = \cos(a)\cos(b) \mp \sin(a)\sin(b)$ and $\sin(a \pm b) = \sin(a)\cos(b) \pm \cos(a)\sin(b)$)

Answer: your answer here.



5. **Matrix Transformation (Equations)** (12 points)

Various conic sections are given in parametric form. Each conic section is then rotated 45° counterclockwise around the origin. Use this information to answer the following questions:

(a) A parabola is given as $x(t) = -t\sqrt{3}, y(t) = -12t^2$.

i. Find the equation of the **parabola** after the transformation. (2 points)

Answer: your answer here.



ii. Find the **focus** of the transformed parabola. (1 point)

Answer: your answer here.



iii. Find the equation of the **directrix** of the transformed parabola. (1 point)

Answer: your answer here.



(b) A circle is given as $x(t) = 3\cos(t), y(t) = 3\sin(t)$.

i. Find the equation of the **circle** after the transformation. (2 points)

Answer: your answer here.



ii. Find the **center** of the transformed circle. (1 point)

Answer: your answer here.



iii. Find the **radius** of the transformed circle. (1 point)

Answer: your answer here.



(c) An ellipse is given as $x(t) = 5\cos(t), y(t) = 3\sin(t)$.

i. Find the equation of the **ellipse** after the transformation. (2 points)

Answer: your answer here.



ii. Find the **foci** of the transformed ellipse. (1 point)

Answer: your answer here.



iii. Find the **major and minor axes** of the transformed ellipse. (1 point)

Answer: your answer here.



(d) A hyperbola is given as $x(t) = 4\sec(t), y(t) = 2\tan(t)$.

i. Find the equation of the **hyperbola** after the transformation. (2 points)

Answer: your answer here.



ii. Find the **foci** of the transformed hyperbola. (1 point)

Answer: your answer here.



iii. Find the equations of the **asymptotes** of the transformed hyperbola. (1 point)

Answer: your answer here.

